

Sanyam Sharma

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WORK EXPERIENCE

Indian Institute of Technology, Hyderabad

Hyderabad, Telangana

Research Intern

Jan 2026 – Present

- Developed an AI-driven network automation system translating natural language intents into SONiC configurations
- Integrated Batfish verification and GNS3 digital-twin testing for automated network change validation
- Built GNS3 REST API-driven workflows for automated network topology discovery and configuration analysis in SONiC environments
- Created Terraform code-generation datasets and benchmarked SLMs using CodeBLEU, CodeBERTScore and Pass@k metrics

Indian Institute of Technology, Mandi

Mandi, Himachal Pradesh

Summer Intern

Jun 2025 – Jul 2025

- Generated speech embeddings using MFCC, wav2vec2.0 and wav2vec2-phoneme for language identification
- Implemented a Conformer-based PyTorch architecture achieving 79% accuracy on unseen multilingual speech samples
- Developed a PyQt6 desktop GUI integrating pretrained AASIST and RawNet models for spoofed speech detection

Indian Institute of Technology, Mandi

Mandi, Himachal Pradesh

Research Intern

Jun 2024 – Jun 2024

- Set up data pipeline using Telegraf and InfluxDB to collect and store sensor data from an ESP32 via MQTT.
- Deployed the system on a Raspberry Pi using Docker for real-time data processing.

EDUCATION

Jawaharlal Nehru Government Engineering College

Mandi, Himachal Pradesh

B.Tech – CSE (AI & ML)

2022 – 2026

Government Senior Secondary School, Kangoo

Hamirpur, Himachal Pradesh

Senior Secondary (Non-Medical)

2022

PROJECTS

AI-Powered Assistive Vision for Blind People - Major Project

Aug 2025 – Dec 2025

Tech: gTTS, FaceNet, Gemini-API, pytesseract, RaspberryPi

- Developed an AI-powered assistive vision smart cap for blind individuals using Raspberry Pi, camera, LiDAR, audio module
- Implemented OCR-based text recognition to read printed and convert it to audio output in real time
- Integrated obstacle detection with audio-based navigation for safe mobility

Extension for malicious QR code and URL detection - Capstone Project

Jan 2025 – May 2025

Tech: Python, Flask, Scikit-Learn, JavaScript

- Engineered features from large-scale malicious and legitimate URL/QR datasets
- Built a stacked ensemble ML model (5 base + meta-classifier), achieving 91.3% accuracy
- Developed a Chrome extension with Flask backend for real-time malicious URL and QR code detection

SpoofedSpeechGUI | GitHub

Tech: Python, PyQt6, PyTorch, Librosa

- Extracted audio features using Librosa and applied pretrained AASIST and RawNet models for spoofed speech detection
- Developed a PyQt6 desktop GUI application for real-time inference

RNN-QA | GitHub | Docker

Tech: Docker, FastAPI, Python, PyTorch

- Trained a single-layer RNN on a question-answer dataset and wrapped model in FastAPI for inference
- Containerized and published a Docker image with **650+ pulls** on Docker Hub

TECHNICAL SKILLS

Languages: Python, C++, SQL, HTML, CSS

Libraries: PyTorch, FastAPI, Scikit-learn, TensorFlow (Keras), NumPy, Pandas, Flask

Tools: Docker, Git, GitHub, Linux, Ollama, Postman, Ansible, Terraform, GNS3, Batfish

PUBLICATIONS

HybridStack-MLP: Advanced Ensemble Learning for Malicious QR Code and URL Detection

2025

- Presented at *2nd IEEE International Conference for Women in Computing (InCoWoCo 2025)*; to appear in IEEE Xplore.
- Ensemble learning model achieving 91.3% accuracy in malicious QR code and phishing URL detection.

ACHIEVEMENTS

Winner — State Level Smart Hackathon (SLSH), Himachal Pradesh

2025